

YU CAO

Computational Linguist

📍 Berlin, Germany

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🐙 [pterosdiacos.github.io](https://github.com/pterosdiacos)

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PROFESSIONAL PROFILE

- Linguistics** Specialized in computational linguistics, semantics, syntax, discourse analysis. Strong critical-thinking and problem-solving skills.
- Computation** Expertise in NLP/NLU modeling and machine learning algorithms. 20+ years of experiences in algorithms and data structures.

EXPERIENCES

Applied Researcher

eBay

📅 11/2021 – present

📍 Berlin, Germany; San Jose, CA

- Work at Memory Science.
- Build grounded AI solutions to power user memory engine.

Ontology Linguist

Facebook via PRO Unlimited

📅 08/2019 – 10/2019

📍 Redmond, WA

- Worked at Assistant Cross-Functional Group.
- Prototyped cross-domain semantic parsing models for question answering.
- Automated inferences in OWL 2 ontology knowledge graphs.

Teaching Assistant

Rutgers University

📅 09/2018 – 08/2020

📍 New Brunswick, NJ

- Courses taught: *Intro to Linguistic Theory*, *Invented Languages*.

Research Assistant

The Chinese University of Hong Kong

📅 12/2014 – 07/2015

📍 Hong Kong, China

- Worked at Center for Sign Linguistics and Deaf Studies.
- Provided linguistic expertise in developing Chinese grammatical knowledge assessment software.

EDUCATION

Rutgers University

Ph.D. in Linguistics

📅 09/2016 – 10/2021

📍 New Brunswick, NJ

The Chinese University of Hong Kong

M.A. in Chinese Linguistics & Language Acquisition

📅 09/2013 – 06/2014

📍 Hong Kong, China

Sichuan Normal University

B.A. in Teaching Chinese as a Second Language

📅 09/2009 – 06/2013

📍 Chengdu, China

PUBLICATIONS

- [Semantic Similarity Based Label Augmentation for Visual Classification](#)
Tiny Papers Track at ICLR 2023.
👤 Yu Cao.
- [Plurality and quantification in graph representation of meaning](#) (🔗 code)
Doctoral dissertation on arXiv:2112.06448.
👤 Yu Cao.
Advisor: Simon Charlow.
- [Automated fact-value distinction in court opinions](#)
European Journal of Law and Economics 2020.
👤 Yu Cao, Daniel Chen & Elliot Ash.
Automated classification of fact vs. value statements in written judicial decisions.
- [Investigating BERT's knowledge of language: Five analysis methods with NPIs](#)
Proceedings of EMNLP 2019.
👤 Alex W., Yu Cao & 14 others.
Supervisor: Sam Bowman.
Role: constructed and conducted the main transfer learning experiments.

SKILLS

Technical

- Python, C, Haskell, SQL, PyTorch, Tensorflow
- Linguistic analysis, symbolic logic, statistics

ML/NLP Coursework

- [Natural language processing](#) 🔄
📅 09/2020 🏛️ HSE@Coursera
- [Bayesian methods for machine learning](#) 🔄
📅 08/2020 🏛️ HSE@Coursera
- [Introduction to deep learning](#) 🔄
📅 07/2020 🏛️ HSE@Coursera
- Machine learning
📅 07/2020 🏛️ Stanford@Coursera
- Seminar: Linguistic knowledge in reusable sentence encoders
📅 05/2019 🏛️ NYU
- Graph formalisms for meaning representations
📅 07/2018 🏛️ NASSLLI@CMU
- NLU & computational semantics
📅 05/2018 🏛️ NYU

Languages

EN (fluent) ZH (native) JA (JLPT N1) DE (A2)